# Local time stepping in the Nitsche solver

How it works, what we built, and when it is stable · revision 1, 19 Oct 2026

**Revision 1.** First version, written for Chunhui Zhao to get a general picture of item 7 of the profiling plan and to read the stability results in full. It covers the phases built so far: the cluster report (P0), the serial multirate stepper with its tests (P1) and damping and absorbing faces per cluster (P2). All of this code sits on `7429df0` of the profiling branch and goes to you for review with this document; it is not used by any run yet, and every existing run is unchanged bit for bit (§6). The design it follows is `lts-kernels-design-rev1.pdf` (cited as "design §n"). Inferences are labelled as such.

## 1. The short version

**The problem.** Today every tet takes the same time step. An explicit step is stable only while it is shorter than the time a wave needs to cross the smallest (or flattest) tet, so a handful of small tets near the fault set the step for all 1.9 million tets of the TPV205 200 m mesh. Most tets could take a step two to eight times longer.

**The idea.** Local time stepping (LTS) sorts the tets into a few *clusters* by the step each one can take on its own, in powers of two: cluster 0 steps at $\Delta t_0$, cluster 1 at $2\Delta t_0$, cluster 2 at $4\Delta t_0$, cluster 3 at $8\Delta t_0$. Each cluster is updated only at its own rate, and after one *sync interval* of $8\Delta t_0$ every tet is back at the same time. With four clusters, TPV205 200 m needs 4.24 times fewer element updates than global stepping (Figure 1), and TPV104 200 m 3.62 times fewer (P0 report). A model of parallel runs that waits for the slowest rank keeps 3.55 times at 2304 ranks, provided the partition balances every cluster on every rank (§3.6).

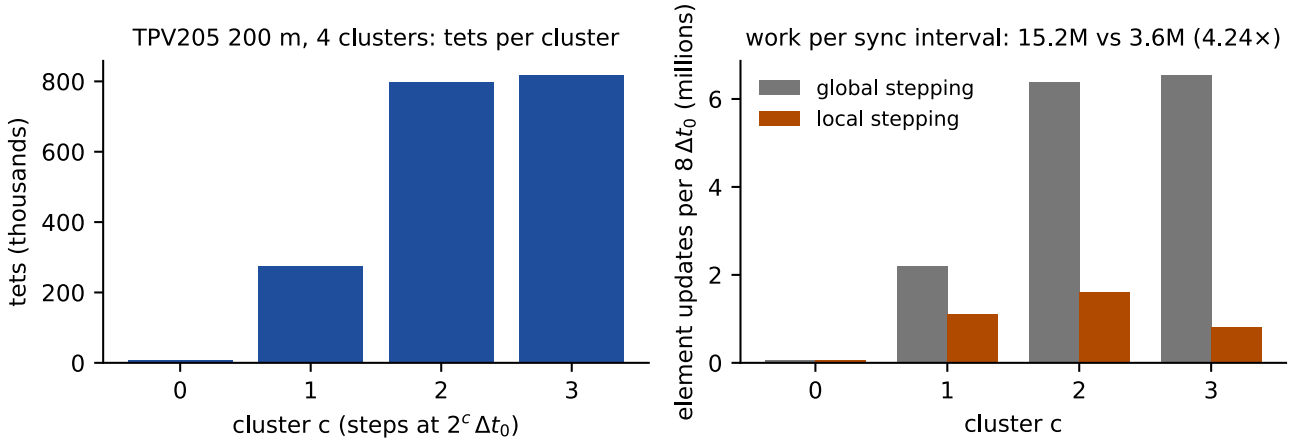



Figure 1. TPV205 200 m with four clusters (the P0 report's rule clustering, clusters 3 to 6 merged into 3). Left: most tets sit in clusters 2 and 3. Right: element updates over one sync interval of $8\Delta t_0$. Global stepping updates every tet 8 times; local stepping updates cluster $c$ only $8/2^c$ times.

**The method.** We use a *multirate leapfrog*: every cluster takes ordinary leapfrog steps of its own length, in physical time, and the force is split so that each tet pushes with its own share at its own rate (§3). With one cluster it is exactly today's central-difference step. The fault will sit inside one cluster and run its existing algorithm at that cluster's step, as SeisSol does (§3.5).

**Stability, in one paragraph.** Global leapfrog is stable for any step up to its limit $\Delta t_{\mathrm{CFL}} = 2/\omega_{\max}$. The multirate leapfrog is not: without damping it is stable only up to about 0.65 of that limit. Above it, a slow cluster's kicks land on the border DOFs every $2\Delta t_0$, which is half the period of the fastest vibrations there, and kicks at that rhythm pump energy into them, the way a child pumps a swing by standing up twice per swing (§5). A one-mode model predicts this band, $1/\sqrt{2(1+\kappa)} < r < 1/\sqrt{2(1-\kappa)}$ with $r = \Delta t_0/\Delta t_{\mathrm{CFL}}$, and puts it where we measured it. Production steps sit at $r = 0.38$ to 0.47, below the band, and you accepted a gate of no growth up to $r = 0.60$. With Kelvin–Voigt damping the band disappears on both test boxes, and the local stepping limit equals the global damped limit at all three damping levels (about 0.77, 0.59 and 0.39 at $\bar{\eta} = 0.05, 0.1, 0.2$; §5.7).

## 2. How a time step works today

The discrete equation of motion is

$$M\ddot{u} + (K_\eta + C)\dot{u} + Ku = F(t) + G\,d + G_\eta V,$$

with $M$ the element-wise mass, $K$ the SIPG stiffness, $K_\eta$ the Kelvin–Voigt damping operator, $C$ the dashpot of the absorbing faces, and $G\,d$, $G_\eta V$ the slip loads of the fault. MFEM's `CentralDifferenceSolver` (NewmarkSolver::Step) advances it. Written as kick–drift–kick, one step is

```
v += dt/2 a                          # kick (half)
u += dt v                            # drift
a  = (M + dt/2 C)^-1 (F + G d - K u - (K_eta + C) v)    # new force, v = mid-step velocity
v += dt/2 a                          # kick (half)
```

Each vibration mode of frequency $\omega$ turns by an angle $\theta$ per step with $\sin(\theta/2) = \omega\Delta t/2$. While $\omega\Delta t \le 2$ the mode only rotates, so its amplitude stays constant; beyond it the amplitude grows every step. The step is therefore limited by the fastest mode, $\Delta t \le \Delta t_{\mathrm{CFL}} = 2/\omega_{\max}$, and $\omega_{\max}$ lives in the smallest or flattest tet: $\omega_{\max,e} \propto p^2 c_p/h_e$ with $h_e = 6V_e/A_e$. The driver uses this estimate, $\Delta t = \mathrm{cfl}\cdot C(p)\min_e h_e/(p^2 c_p)$ with `cfl = 0.5` (ComputeMaxDt, called at driver:1061).

Throughout this document $r = \Delta t_0/\Delta t_{\mathrm{CFL}}$ is the step as a fraction of the true global limit, with $\omega_{\max}$ measured by power iteration on $M^{-1}K$. The h-based estimate is conservative: `cfl = 0.5` gives $r = 0.38$ on the TPV205 5000 m gmsh mesh and $r = 0.47$ on the graded test box (P1 logs). I infer, without measuring, that the 200 m gmsh meshes behave like the 5000 m one.

## 3. How local time stepping works

### 3.1 Clusters

Every tet gets its own stable step $\Delta t_e$ from the same formula as `ComputeMaxDt`. With $\Delta t_0 = \min_e \Delta t_e$, tet $e$ goes into cluster $c_e = \lfloor \log_2(\Delta t_e/\Delta t_0) \rfloor$, so it steps at $2^{c_e}\Delta t_0 \le \Delta t_e$. SeisSol's rules then adjust the bins (BuildLtsClustering, already in the repository for the velocity-stress path):

- **Neighbours differ by at most one level.** A tet whose neighbour is two levels finer is moved down. Each border (*seam*) then joins two rates, one twice the other.
- **Both tets of a fault face share one cluster**, so the fault closure sees one step.

- **At most four clusters.** The top levels merge into one. Four clusters keep 93 to 96% of the ideal gain (4.24 of 4.54 times on TPV205 200 m), and each extra level adds exchange rounds in parallel.

| Mesh (p = 2) | Tets | Ideal, no cap | Cap 4 | Tick model, 2304 ranks, per-cluster partition |
|---|---|---|---|---|
| TPV205 200 m | 1,899,271 | 4.54× | 4.24× | 3.55× |
| TPV104 200 m | 2,463,429 | 3.77× | 3.62× | 3.36× |

The numbers come from the P0 report (`/mnt/project-files/profiling/lts-p0/`), which reproduces the design's tables exactly on all four meshes.

### 3.2 Splitting the force by element

A leapfrog kick needs "the force of cluster $c$". The stiffness is a sum of volume terms, which belong to one tet, and face terms, which couple two tets. A face between a cluster-0 tet and a cluster-1 tet must be split somehow. Giving the whole face to the finer tet looks natural but fails: a SIPG face term on its own is indefinite (a field with a small jump and a large normal stress makes it negative), so the fine cluster's force would contain a coarse tet's face stress without the volume term that keeps it positive, and could create energy.

We split every face term into two halves instead, one per side (design §3.4 step 2, §7.3):

$$a_F(u,w) = a_F^{(1)} + a_F^{(2)}, \qquad a_F^{(i)}(u,w) = -\tfrac{1}{2}\int_F \sigma(u_i)n \cdot [\![w]\!] - \tfrac{1}{2}\int_F \sigma(w_i)n \cdot [\![u]\!] + \pi\,\tfrac{p_i}{4}\int_F [\![u]\!]\cdot[\![w]\!].$$

Side $i$'s half uses only side $i$'s stress and side $i$'s share of the penalty $\pi(p_1+p_2)/4$ that the code already uses. Tet $e$ then owns

$$G_e = \mathrm{vol}_e + \sum_{F \ni e} a_F^{(e)} + \sum_{\text{Dirichlet } F \ni e} a_F, \qquad K = \sum_e G_e, \qquad K^{(c)} = \sum_{e \in c} G_e.$$

Three facts make this the right split. First, $\sum_c K^{(c)} = K$ exactly; the test finds $3 \times 10^{-16}$ relative difference in the largest entry. Second, every $G_e$ is positive semidefinite, so every cluster's force is a proper spring that stores energy; we checked all tets of the boxes and of the TPV205 5000 m gmsh mesh, slivers included, and the smallest eigenvalue is $-1.9 \times 10^{-16}$ of the largest. Third, $G_e$ has rows on the neighbour's face DOFs, so when tet $e$ kicks, it pushes its neighbour back with the opposite force and linear momentum is conserved (inference from the rigid-motion argument of design §3.4). Kelvin–Voigt damping splits the same way, and because every term of $G_e$ uses side $e$'s moduli, $G_e$ of $K_\eta$ is just $\beta_e$ times $G_e$ of $K$. The dashpot block $C_e$ of an absorbing face belongs to its one tet.

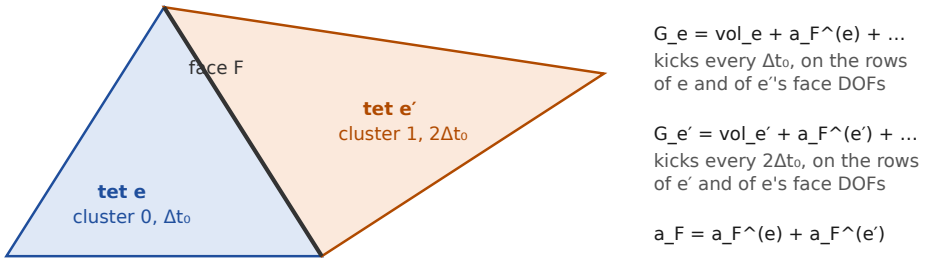



Figure 2. The element-owner split at a seam. Each tet owns its volume term and its half of every face term. The DOFs on face F are kicked by both tets, each at its own rate; that double kicking is where the stability limit of §5 comes from.

### 3.3 One sync interval

With $N_c$ clusters and $L = N_c - 1$, a sync interval has $2^L$ ticks of $\Delta t_0$. Cluster $c$ starts a step at ticks $m$ with $m \bmod 2^c = 0$ and ends one when $(m+1) \bmod 2^c = 0$. At each tick (design §3.4 step 3):

```
for m = 0 .. 2^L - 1:                          # tick m, t = t_S + m dt0
    for c starting a step:  v += dt_c/2 a_c       # start kicks (reuse a_c from the last end)
    u += dt0 v;  t += dt0                          # every DOF drifts at constant velocity
    E = clusters ending a step at t                # 0 .. top
    for c in E:  a_c = B^-1 ( -K^(c) u - K_eta^(c) v - C^(c) v )   # all at the same u, v
    for c in E:  v += dt_c/2 a_c                   # end kicks
# at t_S + 2^L dt0 every cluster has ended a step: v is synchronised
```

Figure 3 shows the schedule for three clusters. A DOF inside cluster 1 is kicked every $2\Delta t_0$ and drifts in between. A DOF on a 0/1 seam is kicked every $\Delta t_0$ by its cluster-0 neighbour and every $2\Delta t_0$ by its own tet. With one cluster the loop is today's kick–drift–kick step; the test finds $2.4 \times 10^{-14}$ relative difference from `CentralDifferenceSolver` after 1000 steps, and $6.4 \times 10^{-14}$ with damping and absorbing faces.

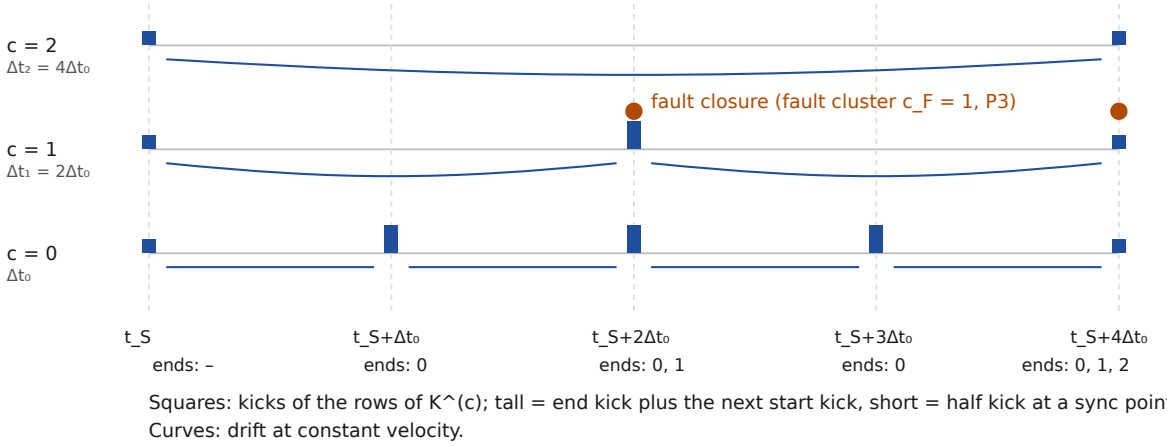



Figure 3. One sync interval with three clusters (design Figure 2). Each cluster kicks at its own instants and drifts in between; all clusters meet at the ends of the interval.

### 3.4 Damping and absorbing faces (P2)

**Kelvin–Voigt.** Today the damping force uses the mid-step velocity $v^* = v^n + \frac{1}{2}\Delta t\, a^n$, lagged by half a step (ImplicitSolve). In the loop above, cluster $c$'s force uses the velocity just before its

end kick, which for its own DOFs is the velocity at the middle of its own step. So each cluster lags its damping by half of its own step, as the design asked (design §3.4 step 5). With `kv_scale = "node"`, $\beta_e \propto h_e/p$, so every cluster has about the same damped stability fraction; the test confirms it (§5.7).

**Absorbing faces.** Today the step solves $(M + \frac{1}{2}\Delta t\, C)\, a = r$ element by element, which makes the dashpot implicit (NitscheMass). With LTS a tet with an absorbing face uses $B_e = M_e + \frac{1}{2}\Delta t_{c_e} C_e$, with its own cluster's step, but only at ticks where its own cluster ends. At ticks where only a faster neighbour ends, the neighbour's kick on that tet's face DOFs uses the plain $M_e$. The first version used $B_e$ at every tick; the order test caught it (error ratios 4.07 then 3.40 instead of 4), and §8.2 shows why it leaves an error of first order. With the fix the ratios are 4.23 and 4.06 at p = 1, and 4.10 and 4.03 at p = 2.

### 3.5 What comes next: fault and parallel runs

**Fault (P3).** All fault faces and both tets of each lie in one cluster $c_F$ (cluster 1 on both 200 m meshes, since the smallest tets are near the fault but rarely on it). `AdvanceFault_` runs unchanged at the ticks where $c_F$ ends a step, with $\Delta t_F = 2^{c_F}\Delta t_0$. Every fault DOF then has its physical displacement, and the velocity before the end kick plays the role of $v^*$. The check $|V|\Delta t/D_c \le 2$ sees twice today's step; if it fails, the fault goes into cluster 0 at a cost of 6% of the ideal gain (design §3.6).

**Parallel (P4).** Each tick needs the ghost values of the clusters that end there. The design merges them into one exchange per tick, posted early and completed late, and needs a partition that balances every cluster on every rank. P0 shows why: MFEM's default cut puts 9 times the mean number of cluster-0 tets on the busiest of 2304 ranks and keeps only 2.14 of the 4.24 times, while the multi-constraint cut (BuildLtsAwarePartition) keeps 3.55 times.

## 4. What we built (code guide)

The flow at run time will be: the driver reads the serial mesh, computes $\Delta t_e$, clusters, partitions, splits the operator per cluster, and steps sync intervals instead of `ode.Step`. P0 to P2 built and tested every piece of that flow except the driver hook-up, the fault and MPI. New files carry no GitHub link because they are not pushed yet; they are in the review diff.

| Function (file) | Inputs → outputs | What it does |
|---|---|---|
| `--lts-report` in the driver; `BuildNitscheLtsMesh`, `PrintNitscheLtsReport` (new `solver/nitsche_lts_setup.hpp`) | serial mesh, p, cfl, c_p, fault attribute → per-tet $h_e$, $\Delta t_e$, faces, fault pairs; a report | P0. Computes $\Delta t_e$ with the driver's formula, clusters with several options (raw, rules, wiggle, caps, fault forced into cluster 0), and models the speedup of two partitions on given rank counts. Prints and exits; no run changes. |
| BuildLtsClustering (existing) | $\Delta t_e$, element graph, fault pairs, options → $c_e$, $\Delta t_0$, cluster count | Rate-2 binning with SeisSol's rules; asserts $2^{c_e}\Delta t_0 \le \Delta t_e$. Used as is. |
| `DGElasticityIPCombinedIntegrator::SetSide(s)` (modified integrator) | side 0, 1 or 2 → the face matrix of the whole form or of one side's half | Masks the consistency, symmetry and penalty terms to side $s$'s half ($a_F^{(s)}$ above). Side 0, the default, is the old form bit for bit. |
| `NitscheLtsSplit` (new `solver/nitsche_lts_stepper.hpp`) | DG space, $\lambda, \mu$, Dirichlet marker, penalty → $K^{(c)}$ per cluster; $C^{(c)}$; $G_e$ on a patch | Assembles $K^{(c)} = \sum_{e \in c} G_e$ as sparse matrices with the same integrators and face loops as the assembled $K$. Built on $\beta_e\lambda_e, \beta_e\mu_e$ it gives $K_\eta^{(c)}$. `AssembleAbsorbingClusters` gives each cluster its tets' dashpot blocks. |
| `NitscheMultirateLeapfrog` (new, same file) | $K^{(c)}$, $K_\eta^{(c)}$, $C^{(c)}$, $M^{-1}$, $B_k^{-1}$, $\Delta t_0$; $(u, v, t) \to (u, v, t)$ one sync interval later | The loop of §3.3. `Init` computes every $a_c$ with $M^{-1}$, as MFEM's first call does; `Step` advances one interval; `StepKicked` is the same map on the state just after the start kicks, used for exact eigenvalues (§8.4). |
| `NitscheMass`, new optional `half_dt_per_elem` (modified) | per-tet $\frac{1}{2}\Delta t \to (M_e + \frac{1}{2}\Delta t_e C_e)^{-1}$ | Lets each absorbing tet use its own cluster's step. Without the argument every bit is as before. |
| `tests/verification/test_nitsche_lts.cpp`, `nitsche_lts_eig.py` (new) | graded boxes or a gmsh mesh → test lines; dense maps → eigenvalues | Tests L-1 to L-10 (§5, §6) and the eigenvalue scan. |

The core of the stepper, as written:

```
Step(u, v, t):                                    # one sync interval
  for m in 0 .. 2^L - 1:
    for c in 0 .. L:   if m % 2^c == 0:  v += 1/2 2^c dt0 a[c]
    u += dt0 v;  t = t0 + (m+1) dt0
    top = largest k with (m+1) % 2^k == 0          # clusters 0 .. top end now
    mass = B[top] (absorbing faces) or M
    for c in 0 .. top:  a[c] = mass^-1 ( -(K[c] u + Keta[c] v + C[c] v) )
    for c in 0 .. top:  v += 1/2 2^c dt0 a[c]
```

`B[k]` holds $M_e + \frac{1}{2}\Delta t_{c_e} C_e$ on the absorbing tets with $c_e \le k$ and $M_e$ elsewhere. The prototype builds one `NitscheMass` per $k$; a production version would apply the dense blocks selectively.

## 5. Stability

### 5.1 What stable means here, and how we test it

The scheme is linear. One sync interval maps the state $z$ (displacements and velocities) to $\Phi z$, and the run is $z_k = \Phi^k z_0$. It is stable when no eigenvalue of $\Phi$ has $|\lambda| > 1$; an eigenvalue with $|\lambda| = 1 + \delta$ multiplies its mode by $e^{\delta k}$ after $k$ intervals. We test it three ways.

1. **Exact spectra on small boxes.** The test program writes $\Phi$ column by column (one interval from each unit vector), and numpy computes all eigenvalues (`nitsche_lts_eig.py`). Round-off puts $\max|\lambda| - 1$ at about $5 \times 10^{-14}$ for a stable map, so the gate is $10^{-10}$. This catches weak instabilities that no time run can see.
2. **Growth on larger meshes.** On the TPV205 5000 m mesh (11,166 tets) $\Phi$ is too large to write out, so the test runs random data and watches the energy grow or decay.
3. **Long energy runs** at the production step, over $10^5$ intervals.

Every test runs global stepping (one cluster) next to local stepping on the same mesh, so a difference is the multirate scheme's.

### 5.2 Why the multirate scheme can be unstable: resonance

Global leapfrog applies every force at every step, and each mode just rotates by $\theta$ per step. In the multirate scheme a seam DOF gets two kinds of kicks: small ones every $\Delta t_0$ from the fast tet and larger ones every $2\Delta t_0$ from the slow tet. For a fast mode on the seam, the slow tet's share of the stiffness is therefore not felt continuously but in bursts, once every $2\Delta t_0$: the spring is switched on and off with that period. A spring modulated at twice the vibration frequency pumps energy into the vibration. This is parametric resonance, the way a child pumps a swing by standing up and crouching twice per swing (the Mathieu equation is the textbook case). Here it happens when the fast mode turns about half a cycle between two slow kicks, $\Theta = 2\theta \approx \pi$. It is the known resonance of impulse multiple-time-step methods in molecular dynamics (r-RESPA; Biesiadecki and Skeel 1993; García-Archilla, Sanz-Serna and Skeel 1998). It has nothing to do with each cluster's own step limit: every cluster alone is stable to 0.92 to 1.4 times its own $2^c \Delta t_{\mathrm{CFL}}$ (P1, "cluster limits").

### 5.3 A one-mode model that predicts the band

Take one mode of frequency $\omega$ whose stiffness comes from two clusters: a share $1 - \kappa$ is kicked every $\Delta t_0$, a share $\kappa$ every $2\Delta t_0$. For a mode inside cluster 0, $\kappa = 0$; for a mode with weight on seam DOFs, $\kappa$ is the part of its stiffness that the slow tet's $G_e$ provides. One sync interval is two fast leapfrog steps followed by the slow kick (§8.3).

> **Result 1 (one-mode model).** With $r = \Delta t_0\, \omega/2$ the step as a fraction of this mode's own global limit, the sync-interval map has $|\lambda| > 1$ exactly when
>
> $$\frac{1}{\sqrt{2(1+\kappa)}} \;<\; r \;<\; \frac{1}{\sqrt{2(1-\kappa)}},$$
>
> a band around $r = 1/\sqrt{2} \approx 0.707$, where the fast leapfrog turns the mode by exactly half a cycle in $2\Delta t_0$. Below the band and above it (up to $r = 1$) the map is stable. Since $0 \le \kappa \le 1$, the band never reaches below $r = 1/2$.

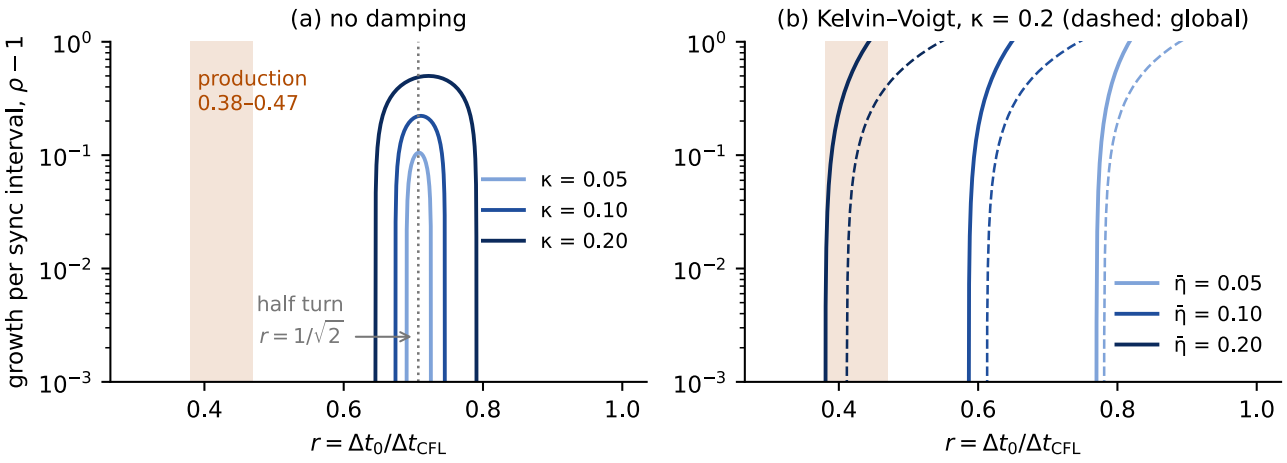



Figure 4. The one-mode model. (a) Without damping, growth per sync interval against $r$ for three coupling shares $\kappa$: one band just around the half turn, wider for stronger coupling. The shaded strip is the production range. (b) With Kelvin–Voigt damping ($\kappa = 0.2$, the slow part damped with twice the fast part's $\beta$, as for a tet twice the size), solid local, dashed global: the band is gone and the limit is the damped limit, lowered by at most 0.03.

Two things follow, both inferences from the model. First, a mode slower than $\omega_{\max}$ has its band at a larger $r$ (its own $r$ is $r\,\omega/\omega_{\max}$), so modes across the spectrum fill the range from the first band up to 1. Second, the first band belongs to the fastest modes that touch a seam, and starts at $r = 1/\sqrt{2(1+\kappa)}$: 0.67 for $\kappa = 0.1$, 0.65 for $\kappa = 0.2$. Which modes those are depends on how close the smallest tets sit to a cluster border.

### 5.4 What we measured without damping (P1)

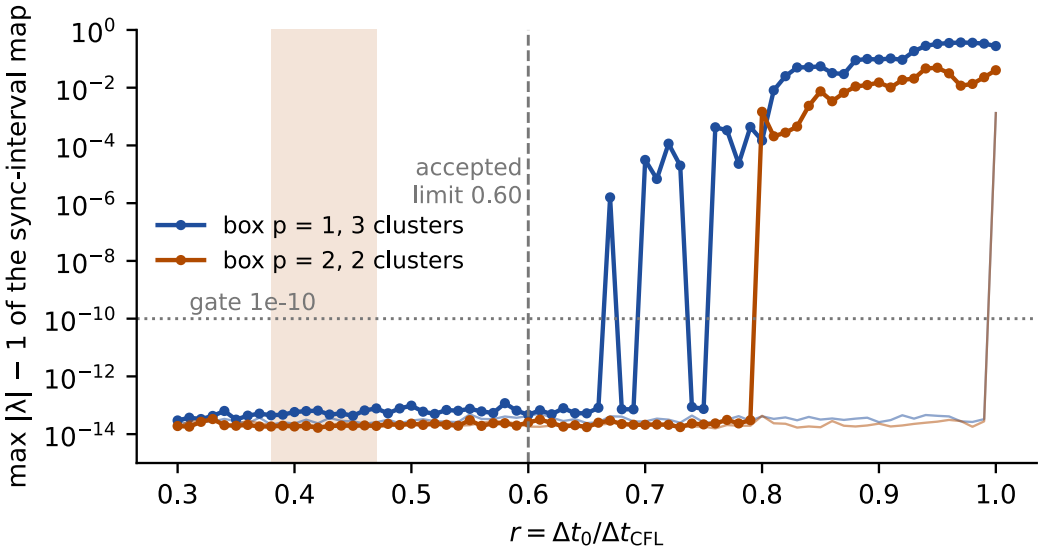



Figure 5. Exact spectra of the sync-interval map on the graded boxes, $\max|\lambda| - 1$ against $r$ (points: clusters; faint line: one cluster on the same box). The one-cluster map stays at round-off below $r = 1$ (at $r = 1$ itself, the global limit, it reaches $10^{-3}$ on the p = 2 box). The p = 1 box is at round-off up to 0.66 and grows above; the p = 2 box is at round-off up to 0.79 and grows from 0.80.

| Test | Stable up to | Above it |
|---|---|---|
| Box p = 1, 162 tets, 3 clusters, exact spectra | 0.66 ($\le 8 \times 10^{-14}$) | weak bands at 0.67 ($1.6 \times 10^{-6}$) and 0.70–0.80 (up to $4.3 \times 10^{-4}$); strong from 0.81 ($8 \times 10^{-3}$ to 0.37) |
| Box p = 2, 48 tets, 2 clusters, exact spectra | 0.79 ($\le 3.1 \times 10^{-14}$) | growth from 0.80 ($1.5 \times 10^{-3}$), up to $4.9 \times 10^{-2}$ by 0.95; no weak band below 0.80 |
| Box p = 1 and p = 2, energy growth, 2000 intervals | 0.80 and 0.81 | growth from 0.81 and 0.82 (the weak bands are too slow to see) |
| TPV205 5000 m gmsh mesh p = 1, 11,166 tets, 3 clusters, energy growth | 0.65 | growth from 0.70 |
| Box p = 1, two clusters with the seam moved away from the fastest tets | 0.99 | only the global limit at 1.00 |
| Same, seam next to the fastest tets | 0.80 | growth from 0.81 |

The measurements fit the model. The first band starts at 0.67 on the box and between 0.65 and 0.70 on the gmsh mesh, which the model places at $\kappa \approx 0.1$ to 0.2. The p = 2 box has no band below 0.80, above the model's centre $1/\sqrt{2}$: there the fastest modes that touch the seam are slower than $\omega_{\max}$, about $0.88\,\omega_{\max}$ (inference, $0.707/0.80$), so their band sits higher, as §5.3 says. Moving the seam away from the fastest tets, so that no fast mode has weight on a seam ($\kappa \to 0$), removes the instability below the global limit. (The `--seam` mode prints a "predicted" onset of $\frac{\pi}{4}$ times the seam tet's $\Delta t_e$; that older estimate ignores the leapfrog's own phase and the fact that a mode spreads over several tets, and is superseded by Result 1.)

### 5.5 How fast does a weak band grow?

A growth of $\delta$ per interval multiplies a mode by $e$ after $1/\delta$ intervals. TPV205 200 m at p = 2 takes 8,636 steps per simulated second, so with four clusters about 1,080 intervals per second. A band of $4 \times 10^{-4}$ per interval then grows a mode 75-fold in 10 s of simulated time, and $1.6 \times 10^{-6}$ grows it by 2%. Weak bands are slow, but a long run would find them, so the gate keeps the step out of every band: no growth ($|\lambda| \le 1 + 10^{-10}$) up to $r = 0.60$, the value you accepted. That leaves a margin of 0.05 to 0.07 to the first band seen.

### 5.6 Energy over long runs at the production step

The base suite runs $10^4$ intervals at $r = 0.5$: the energy at sync points stays within $[0.998, 1]$ of its start for smooth data on the p = 1 box and within $[0.9997, 1]$ on the p = 2 box. Runs of $10^5$ intervals at $r = 0.38$ and 0.5 on the TPV205 5000 m mesh are in progress (§9).

### 5.7 With damping and absorbing faces (P2)

Kelvin–Voigt damping lowers the global limit to the fraction $\sqrt{1+\zeta^2} - \zeta$ of the undamped one, with $\zeta = \frac{1}{2}\beta\omega_{\max}$: 0.78, 0.61 and 0.41 at $\bar{\eta} = 0.05, 0.1, 0.2$ (KV derivation rev 2.1 §6). The model of Figure 4b predicts that damping also removes the resonance band, so the local stepping limit becomes the damped limit, a little lower when slow tets have the larger $\beta_e$.

The box tests (Kelvin–Voigt in every tet, absorbing faces at $x = \pm L$, growth rate over 3000 intervals, $r$ in steps of 0.02) agree:

| **Box** | $\bar{\eta}$ | **Damped fraction, whole / per cluster** | **First unstable $r$, global / local** | **KV rev 2.1** |
|---|---|---|---|---|
| p = 1, 3 clusters | 0.05 | 0.767 / 0.768–0.787 | 0.80 / 0.80 | 0.78 |
| p = 1 | 0.1 | 0.598 / 0.599–0.627 | 0.62 / 0.62 | 0.61 |
| p = 1 | 0.2 | 0.394 / 0.395–0.424 | 0.42 / 0.42 | 0.41 |
| p = 2, 2 clusters | 0.05 | 0.758 / 0.759–0.775 | 0.78 / 0.78 | — |
| p = 2 | 0.1 | 0.586 / 0.586–0.610 | 0.60 / 0.60 | — |
| p = 2 | 0.2 | 0.381 / 0.382–0.405 | 0.40 / 0.40 | — |

Three points stand out. First, every cluster has the same damped fraction to within 0.03, as the design inferred for `kv_scale = "node"` (design §3.4 step 5). Second, local stepping becomes unstable at the same $r$ as global stepping at every damping level. Third, at $\bar{\eta} = 0.05$ local stepping on the p = 1 box is stable to $r = 0.78$, inside the undamped band of the same box (0.67 to 0.80), so damping did remove it, as the model predicts. The growth-rate test cannot see a band weaker than about $10^{-6}$ per interval, so we also computed the exact spectrum of the damped map on the p = 1 box at $\bar{\eta} = 0.05$. From $r = 0.55$ to 0.78, across the whole undamped band, every eigenvalue has $|\lambda| \leq 1 - 5.2 \times 10^{-3}$ per interval. The least damped mode also decays at the same rate per unit time as with global stepping: one interval is four steps of $\Delta t_0$, and $1 - |\lambda|$ is four times the global step's value to within 0.2% at every $r$. At $r = 0.79$ the global and the local map become unstable together, at the damped limit.

Absorbing faces do not change the limit, because the dashpot is implicit per tet. Over 30 s on the boxes, with no Kelvin–Voigt, local stepping absorbs 0.9652 of the initial energy against 0.9651 for global stepping at p = 1, and 0.9099 for both at p = 2; the energy histories differ by at most 0.0015 of the initial energy at p = 1 and 0.0002 at p = 2.

### 5.8 What it means in practice

- **Production is inside the stable range.** With `cfl = 0.5` the step is $r = 0.38$ to 0.47, below the undamped band (from 0.65) and below the damped limit at $\bar{\eta} \leq 0.1$ (0.59 to 0.60 on the boxes). At $\bar{\eta} = 0.2$ the box's limit is 0.40 to 0.42 for global stepping too, so `cfl = 0.5` is too large there whether LTS is used or not (the KV note's bound is $\bar{\eta} \lesssim 0.15$).
- **A guard in the driver.** The run does not know $r$ exactly, but the h-based estimate is conservative ($r \approx 0.75$ to 0.94 times `cfl` on our meshes). I propose that LTS refuses `cfl > 0.6`, which keeps $r \leq 0.57$ (inference from those two ratios).
- **If a larger step is ever needed.** Three remedies exist, none built: a seam buffer in the clustering that keeps the fastest tets' neighbours in their cluster (lowers $\kappa$, costs some gain); a mollified slow kick (García-Archilla, Sanz-Serna and Skeel 1998); or the Diaz–Grote leapfrog LTS, which has an energy proof but was ruled out for the fault (design §3.3).

## 6. Test results so far

| Test | What it checks | Result |
|---|---|---|
| L-1 split | $\sum_c K^{(c)} = K$ | pass, $\leq 3.3 \times 10^{-16}$ (boxes, TPV205 5000 m) |
| L-2 (b) positivity | every $G_e$ positive semidefinite | pass on every tet, slivers included |
| L-3 (a) one cluster | equals `CentralDifferenceSolver` | pass, $2.4 \times 10^{-14}$ after 1000 steps |
| L-4 (c) growth scan | no growth for $r \leq 0.60$ (the design asked for $r \leq 1$) | pass; growth from 0.81 (p = 1) and 0.82 (p = 2) |
| L-5 (d) energy | bounded at sync points for $r \leq 0.60$ | pass at $r = 0.5$ and 0.6; $r = 0.9$ grows, printed only |
| L-6 (e) order | second order in time | pass, ratios 4.04 to 4.22 |
| L-7 P2 split, (b) damping power | $\sum_c K_\eta^{(c)} = K_\eta$, $\sum_c C^{(c)} = C$, every cluster's damping dissipates | pass ($\leq 6.7 \times 10^{-16}$; all quadratic forms positive; all $C_e$ PSD) |
| L-8 (a) one cluster, damped | equals the damped `NitscheWaveOperator` step | pass, $6.4 \times 10^{-14}$ (p = 1), $2.8 \times 10^{-14}$ (p = 2) |
| L-9 (a) damped limit | cluster fractions agree; local limit $\geq \min(\text{global}, 0.60)$ | pass (§5.7) |
| L-10 (c) absorbing | absorbed energy as global stepping; second order | pass, within 0.0015 of the initial energy; ratios 4.03 to 4.23 after the fix of §3.4 |
| Existing runs | the driver against `7429df0`: TPV205 (p = 1 damped and undamped, p = 2), TPV104 and TPV102 (p = 1, 2) on 4 ranks, TPV205 p = 1 on 1 rank; the six Nitsche gate test sets | every station file byte-identical and every fault field bitwise equal in all 8 runs; gates 74, 22, 49, 13, 9, 42 passed, none failed |

The design's gate (c), stable to $r = 1$, is replaced by "stable to $r = 0.60$" after your decision of 19 Oct; L-4, L-5 and L-9 now test that value. The whole suite, L-1 to L-10 at p = 1 and p = 2, passes: 24 of 24.

## 7. Phases and what is next

| Phase | Content | Status |
|---|---|---|
| P0 | `--lts-report`: clusters, histograms, partition model | done; matches the design on four meshes |
| P1 | serial multirate leapfrog, pure elastic | done; limit 0.65 found, gate 0.60 accepted |
| P2 | Kelvin–Voigt and absorbing faces per cluster | done; all tests pass on the boxes, damped spectra show no band |
| P3 | fault in one cluster; `AdvanceFault_` at $\Delta t_F$; stations at fault instants | next |
| P4 | MPI: per-cluster partition, one exchange per tick | after P3 |
| P5 | TPV205 and TPV104 200 m on Vista; new golds | after P4; results change by the time-discretisation error |

Two decisions from the design remain open before P4: whether mesh optimisation (item 2) comes first, since it removes part of the gain, and whether to keep the cap of four clusters.

## 8. Derivations

### 8.1 One cluster is central difference

With one cluster each tick is: $v \mathrel{+}= \frac{1}{2}\Delta t\, a$; $u \mathrel{+}= \Delta t\, v$; $a = B^{-1}r(u, v)$; $v \mathrel{+}= \frac{1}{2}\Delta t\, a$. MFEM's Newmark step with $\beta = 0, \gamma = \frac{1}{2}$ computes $u \mathrel{+}= \Delta t\, v + \frac{1}{2}\Delta t^2 a$, $v^* = v + \frac{1}{2}\Delta t\, a$, solves for $a$ at $(u, v^*)$, and sets $v = v^* + \frac{1}{2}\Delta t\, a$. Both produce the same numbers in a different order, so they agree to round-off. The first acceleration uses $M^{-1}$ in both (`Init`, and Newmark's first `Mult`). ∎

### 8.2 The dashpot with per-tet steps

Let tet $e$ have an absorbing face and lie in cluster $c_e$, with $h = \frac{1}{2}\Delta t_{c_e}$ and $B_e = M_e + hC_e$. Today's step makes the dashpot implicit: $M(v^+ - v^-) = \frac{1}{2}\Delta t\,(R - Cv^+)$ at the end kick, where $R$ is the elastic force.

**Step 1, a tick where $c_e$ ends.** All kicks on $e$'s rows use $B_e^{-1}$, and only cluster $c_e$'s force contains $-C_e v^-$. Summing the kicks, $B_e(v^+ - v^-) = \sum_c \frac{1}{2}\Delta t_c R_c - hC_e v^-$, so $M_e(v^+ - v^-) = \sum_c \frac{1}{2}\Delta t_c R_c - hC_e v^+$: today's implicit dashpot. The next start kicks reuse the same $a_c$, so they add $M_e^{-1}(\sum_c \frac{1}{2}\Delta t_c R_c - hC_e v^+)$, and the dashpot impulse over a step of $c_e$ is the trapezoid rule on $C_e v$ at its two ends, which is second order.

**Step 2, a tick where only faster clusters end.** Their force on $e$'s face DOFs is elastic, $\frac{1}{2}\Delta t_0 R_0$. With $M_e^{-1}$ it is applied as it is. With $B_e^{-1}$ it would satisfy $M_e\,\delta v = \frac{1}{2}\Delta t_0 R_0 - hC_e\,\delta v$: an extra impulse $-hC_e\delta v = O(\Delta t^2)$ per tick. Summed over the $O(1/\Delta t)$ ticks of a run it is an $O(\Delta t)$ error, which is what the order test saw before the fix (error ratios 4.07, then 3.40 on the p = 1 box and 3.88, 3.33 on the p = 2 box). ∎

### 8.3 The one-mode model (Result 1)

**Assumptions.** (M1) One DOF, $\ddot{x} = -(\omega_f^2 + \omega_s^2)x$, with $\omega^2 = \omega_f^2 + \omega_s^2$ and $\kappa = \omega_s^2/\omega^2$. (M2) The fast force kicks every $h = \Delta t_0$, the slow force every $2h$, in the loop of §3.3. (M3) No damping.

**Step 1, the map.** Kicks only change $v$ by a multiple of $x$, so they commute. Starting just after the start kicks, the interval is a drift, a fast kick $h$ (end plus next start), a drift, and a combined kick of fast $h$ and slow $2h$. With $K_f(h) = \begin{pmatrix} 1 & 0 \\ -h\omega_f^2 & 1 \end{pmatrix}$, $D(h) = \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$ and the fast leapfrog step $L = K_f(h)D(h)$,

$$\Phi = K_s(2h)\, L^2, \qquad \det \Phi = 1.$$

**Step 2, the trace.** $\operatorname{tr} L = 2 - h^2\omega_f^2 = 2\cos\theta$, where $\theta$ is the fast step's rotation angle. Cayley–Hamilton gives $L^2 = 2\cos\theta\, L - I$, so $(L^2)_{12} = 2h\cos\theta$ and

$$\operatorname{tr}\Phi = \operatorname{tr} L^2 - 2h\omega_s^2 (L^2)_{12} = 2\cos 2\theta - 4q\cos\theta, \qquad q = h^2\omega_s^2.$$

**Step 3, the band.** With $\det\Phi = 1$, the map is unstable exactly when $|\operatorname{tr}\Phi| > 2$. Write $c = \cos\theta$. Then $\operatorname{tr}\Phi < -2$ means $4c^2 - 4qc < 0$, that is $0 < c < q$; the other side, $\operatorname{tr}\Phi > 2$, needs $c < (q - \sqrt{q^2+4})/2$, which lies beyond the fast step's own limit for $r \le 1$.

**Step 4, in terms of r.** With $r = h\omega/2$, $c = 1 - \frac{1}{2}h^2\omega_f^2 = 1 - 2r^2(1-\kappa)$ and $q = 4r^2\kappa$. The condition $c > 0$ gives $r < 1/\sqrt{2(1-\kappa)}$, and $c < q$ gives $1 < 2r^2(1+\kappa)$, that is $r > 1/\sqrt{2(1+\kappa)}$. ∎

A numerical sweep of the model (`onedof.py`) gives the same edges: 0.675 to 0.745 for $\kappa = 0.1$ and 0.646 to 0.790 for $\kappa = 0.2$. With damping there is no closed form; `onedof_kv.py` evaluates the same map with Kelvin–Voigt on both parts, lagged as in §3.4, and gives Figure 4b.

### 8.4 Why the eigenvalues use the state after the start kicks

Without damping, $a_c$ depends on $u$ only, so the map from $(u, v)$ at one sync point to the next is well defined. With damping, $a_c$ depends on the velocity before the end kick, which cannot be read off the synchronised $v$, so $(u, v)$ does not close the recursion. The state just after the start kicks does: every acceleration the rest of the interval needs is computed inside it. `StepKicked` maps that state to the same point of the next interval. Without damping its map is similar to the old one, so the spectra agree.

## 9. Where it lives

- **Code**: on `7429df0` of `claude/nitsche-profiling-w336yq`, in the review sent with this document. Modified: `drivers/nitsche_dyn_driver.cpp`, `Makefile`, `solver/nitsche_cfl.hpp`, `integrator/dg_elasticity_ip_combined_integrator.hpp`, `domain/nitsche_mass.hpp`. New: `solver/nitsche_lts_setup.hpp`, `solver/nitsche_lts_stepper.hpp`, `tests/verification/test_nitsche_lts.cpp`, `tests/verification/nitsche_lts_eig.py`.
- **Logs**: `/mnt/project-files/profiling/lts-p0/` (cluster reports), `lts-p1/` (base suite, eigenvalue scans, TPV205 5000 m, seam runs), `lts-p2/` (full suite, p = 2 and damped spectra, gate tests).
- **This document**: `/mnt/project-files/profiling/lts-explainer/`, with the figure and model scripts (`figs.py`, `onedof.py`, `onedof_kv.py`).
- **Still running** when this revision was written: the $10^5$-interval energy runs on TPV205 5000 m (§5.6); their results go to `lts-p2/README.md`.

Biesiadecki, J. J. and Skeel, R. D. (1993), J. Comput. Phys. 109, 318–328. García-Archilla, B., Sanz-Serna, J. M. and Skeel, R. D. (1998), SIAM J. Sci. Comput. 20, 930–963. Tuckerman, M., Berne, B. J. and Martyna, G. J. (1992), J. Chem. Phys. 97, 1990–2001. Diaz, J. and Grote, M. J. (2009), SIAM J. Sci. Comput. 31, 1985–2014.